

Лабораторная работа №1

Тема: Освоение инструментария для выполнения работ, построение простой сети.

2) Создать простейшую сеть, состоящую из 1 коммутатора и 2 компьютеров, назначить им произвольные ip адреса из одной сети

```
PC1> ip 192.168.1.10/24  
Checking for duplicate address...  
PC1 : 192.168.1.10 255.255.255.0
```

```
PC2> ip 192.168.1.20/24  
Checking for duplicate address...  
PC2 : 192.168.1.20 255.255.255.0
```

3) Запустить симуляцию, выполнить команду ping с одного из компьютеров, используя ip адрес второго компьютера

```
PC1> ping 192.168.1.20  
84 bytes from 192.168.1.20 icmp_seq=1 ttl=64 time=0.170 ms  
84 bytes from 192.168.1.20 icmp_seq=2 ttl=64 time=0.273 ms  
84 bytes from 192.168.1.20 icmp_seq=3 ttl=64 time=0.263 ms  
84 bytes from 192.168.1.20 icmp_seq=4 ttl=64 time=1.217 ms  
84 bytes from 192.168.1.20 icmp_seq=5 ttl=64 time=0.195 ms  
PC1>
```

4) Перехватить трафик протокола arp на всех линках(nb!), задокументировать и проанализировать заголовки пакетов в программе Wireshark, для фильтрации трафика, относящегося к указанному протоколу использовать фильтры Wireshark.

PC1 to Switch:

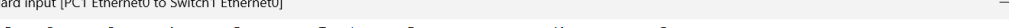
```
Destination: Broadcast (ff:ff:ff:ff:ff:ff) - широковещательный MAC-адрес  
Source: Private_66:68:01 (00:50:79:66:68:01) - MAC-адрес отправителя (источник)  
Type: ARP (0x0806)  
Protocol type: IPv4 (0x0800)
```

Sender MAC address: Private_66:68:01 (00:50:79:66:68:01)

Sender IP address: 192.168.1.10

Target IP address: 192.168.1.20

Устройство с IP 192.168.1.10 спрашивает у IP-адреса 192.168.1.20 какой у него MAC-адрес.



The screenshot shows the Wireshark interface with a packet capture of ARP requests. The packet list pane shows two packets:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Private_66:68:01	Broadcast	ARP	64	who has 192.168.1.20? Tell 192.168.1.10
2	0.000218	Private_66:68:00	Private_66:68:01	ARP	64	192.168.1.20 is at 00:50:79:66:68:00

The packet details pane shows the selected packet (packet 2) with the following structure:

```

Ethernet II, Src: Private_66:68:00, Dst: Private_66:68:01, Type: ARP (0x0806)
Internet Protocol Version 4, Src: 192.168.1.20, Dst: 192.168.1.10
Address Resolution Protocol, Operation: Request, Sender Hardware Address: Private_66:68:00, Target Hardware Address: Private_66:68:01, Target IP Address: 192.168.1.20
  
```

> Frame 1: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface 0	0000	ff ff ff ff ff ff ff 00	79 66 68 01 08 06 00 01	 P yfh
> Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff)	0010	08 00 06 04 00 01 50 50	79 66 68 01 c0 a8 01 0a	 P yfh
> Internet Protocol Version 4, Src: 10.0.0.1, Dst: 255.255.255.255	0020	ff ff ff ff ff ff c0 a8	01 14 00 00 00 00 00 00	
> Address Resolution Protocol (request)	0030	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	

PC2 to Switch:

Sender IP address: 192.168.1.20

Sender MAC address: 00:50:79:66:68:00

Устройство с IP 192.168.1.20 сообщает свой MAC-адрес:
00:50:79:66:68:00.

Target IP address: 192.168.1.10

Target MAC address: 00:50:79:66:68:01

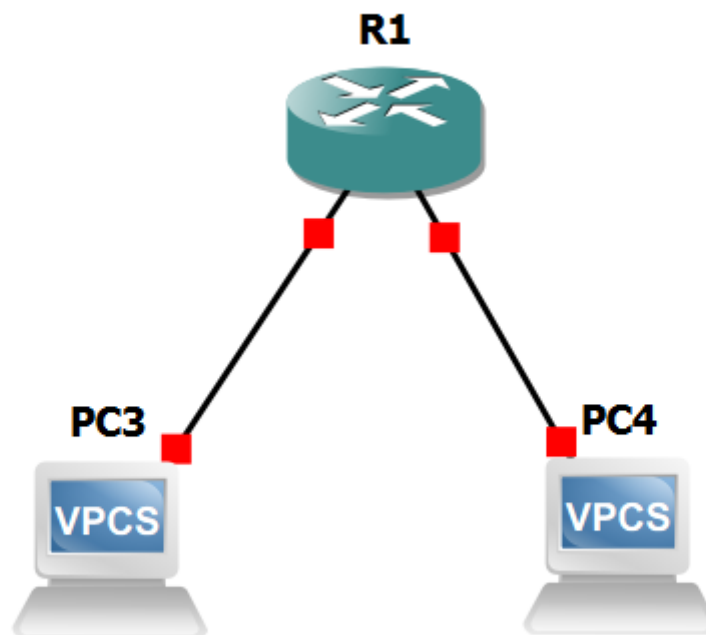
Ответ направляется обратно устройству, которое задавало вопрос.

The screenshot shows the Wireshark network protocol analyzer interface. The title bar reads "Захват с Standard input [PC2 Ethernet0 to Switch1 Ethernet1]". The menu bar includes "Файл", "Правка", "Вид", "Запуск", "Захват", "Анализ", "Статистика", "Телефония", "Беспроводная связь", "Инструменты", and "Справка". The toolbar contains various icons for file operations, packet selection, and analysis. The packet list pane on the left shows a single packet selected, with the filter "arp" entered in the text box. The packet details pane on the right shows the structure of an ARP request:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Private_66:68:01	Broadcast	ARP	64	who has 192.168.1.20? Tell 192.168.1.10
2	0.000187	Private_66:68:00	Private_66:68:01	ARP	64	192.168.1.20 is at 00:50:79:66:68:00

> Frame 1: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits)	0000	ff ff ff ff ff ff ff 00	50 79 66 68 01 08 06 00	01	 P yfh
> Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Broadcast (f	0010	08 00 06 04 00 01 00 50	79 66 68 01 c0 a8 01 0a		 P yfh
> Address Resolution Protocol (request)	0020	ff ff ff ff ff ff c0 a8	01 14 00 00 00 00 00 00		
	0030	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00		

5) Создать простейшую сеть, состоящую из 1 маршрутизатора и 2 компьютеров, назначить им произвольные ip адреса из разных сетей.



Сначала настроим маршрутизатор:

```
R1#enable
```

```
R1#configure terminal
```

```
R1(config)#interface FastEthernet0
```

```
R1(config)#interface FastEthernet0/0
```

```
R1(config-if)#ip address 192.168.3.1 255.255.255.0
```

```
R1(config-if)#no shutdown
```

```
R1(config-if)#exit
```

```

R1#show ip interface
FastEthernet0/0 is administratively down, line protocol is down
  Internet protocol processing disabled
FastEthernet1/0 is administratively down, line protocol is down
  Internet protocol processing disabled
Ethernet2/0 is administratively down, line protocol is down
  Internet protocol processing disabled
Ethernet2/1 is administratively down, line protocol is down
  Internet protocol processing disabled
Ethernet2/2 is administratively down, line protocol is down
  Internet protocol processing disabled
Ethernet2/3 is administratively down, line protocol is down
  Internet protocol processing disabled
Serial3/0 is administratively down, line protocol is down
  Internet protocol processing disabled
Serial3/1 is administratively down, line protocol is down
  Internet protocol processing disabled
Serial3/2 is administratively down, line protocol is down
  Internet protocol processing disabled
Serial3/3 is administratively down, line protocol is down
  Internet protocol processing disabled
R1#enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#inte
R1(config)#interface Fast
R1(config)#interface FastEthernet0
R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 192.168.3.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar  1 00:15:50.271: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar  1 00:15:51.271: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up

```

R1#configure terminal

R1(config)#interface FastEthernet1/0

R1(config-if)#ip address 192.168.4.1 255.255.255.0

R1(config-if)#no shutdown

R1(config-if)#exit

```

R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface FastEthernet1/0
R1(config-if)#ip address 192.168.4.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar  1 00:19:32.187: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state t
o up
*Mar  1 00:19:33.187: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et1/0, changed state to up
R1(config)#
R1#
*Mar  1 00:19:54.955: %SYS-5-CONFIG_I: Configured from console by console
R1#

```

Теперь настроим pc3 и pc4:

PC3> ip 192.168.3.30/24 192.168.3.1

PC3 : 192.168.3.30 255.255.255.0 gateway 192.168.3.1

PC4> ip 192.168.4.40/24 192.168.4.1

PC4 : 192.168.4.40 255.255.255.0 gateway 192.168.4.1

PC3> ip 192.168.3.30/24 192.168.3.1 Checking for duplicate address... PC3 : 192.168.3.30 255.255.255.0 gateway 192.168.3.1	PC4> ip 192.168.4.40/24 192.168.4.1 Checking for duplicate address... PC4 : 192.168.4.40 255.255.255.0 gateway 192.168.4.1
--	--

6) Запустить симуляцию, выполнить команду ping с одного из компьютеров, используя ip адрес второго компьютера

PC3> ping 192.168.4.10

```
PC3> ping 192.168.4.40

192.168.4.40 icmp_seq=1 timeout
84 bytes from 192.168.4.40 icmp_seq=2 ttl=63 time=11.815 ms
84 bytes from 192.168.4.40 icmp_seq=3 ttl=63 time=15.401 ms
84 bytes from 192.168.4.40 icmp_seq=4 ttl=63 time=15.955 ms
84 bytes from 192.168.4.40 icmp_seq=5 ttl=63 time=15.063 ms
```

7) Перехватить трафик протокола arp и icmp на всех линках(nb!), задокументировать и проанализировать заголовки пакетов в программе Wireshark, для фильтрации трафика, относящегося к указанному протоколу использовать фильтры Wireshark

PC3 to R1:

PC3 видит, что IP-адрес назначения находится в другой сети, поэтому пакет необходимо передать через шлюз по умолчанию. Однако PC3 не знает MAC-адрес своего шлюза, поэтому отправляет широковещательный ARP-запрос в локальную сеть, чтобы узнать его MAC-адрес:

No.	Time	Source	Destination	Protocol	Length	Info
13	94.494516	Private_66:68:02	Broadcast	ARP	64	Who has 192.168.3.1? Tell 192.168.3.30
14	94.497768	cc:01:48:2d:00:00	Private_66:68:02	ARP	60	192.168.3.1 is at cc:01:48:2d:00:00
15	94.498732	192.168.3.30	192.168.4.40	ICMP	98	Echo (ping) request id=0x779b, seq=1/256, ttl=64 (reply in 16)
16	94.528006	192.168.4.40	192.168.3.30	ICMP	98	Echo (ping) reply id=0x779b, seq=1/256, ttl=63 (request in 15)
17	95.528532	192.168.3.30	192.168.4.40	ICMP	98	Echo (ping) request id=0x789b, seq=2/512, ttl=64 (reply in 18)
18	95.544180	192.168.4.40	192.168.3.30	ICMP	98	Echo (ping) reply id=0x789b, seq=2/512, ttl=63 (request in 17)
19	96.544539	192.168.3.30	192.168.4.40	ICMP	98	Echo (ping) request id=0x799b, seq=3/768, ttl=64 (reply in 20)
20	96.560446	192.168.4.40	192.168.3.30	ICMP	98	Echo (ping) reply id=0x799b, seq=3/768, ttl=63 (request in 19)
21	97.561526	192.168.3.30	192.168.4.40	ICMP	98	Echo (ping) request id=0x7a9b, seq=4/1024, ttl=64 (reply in 22)
22	97.577397	192.168.4.40	192.168.3.30	ICMP	98	Echo (ping) reply id=0x7a9b, seq=4/1024, ttl=63 (request in 21)
23	98.578609	192.168.3.30	192.168.4.40	ICMP	98	Echo (ping) request id=0x7b9b, seq=5/1280, ttl=64 (reply in 24)
24	98.593253	192.168.4.40	192.168.3.30	ICMP	98	Echo (ping) reply id=0x7b9b, seq=5/1280, ttl=63 (request in 23)

> Frame 13: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bit)

> Ethernet II, Src: Private_66:68:02 (00:50:79:66:68:02), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

▼ Address Resolution Protocol (request)

- Hardware type: Ethernet (1)
- Protocol type: IPv4 (0x0800)
- Hardware size: 6
- Protocol size: 4
- Opcode: request (1)
- Sender MAC address: Private_66:68:02 (00:50:79:66:68:02)
- Sender IP address: 192.168.3.30
- Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
- Target IP address: 192.168.3.1

Маршрутизатор получает этот ARP-запрос и отправляет ARP Reply, сообщая PC3 свой MAC-адрес:

No.	Time	Source	Destination	Protocol	Length	Info
13	94.494516	Private_66:68:02	Broadcast	ARP	64	Who has 192.168.3.1? Tell 192.168.3.30
14	94.497768	cc:01:48:2d:00:00	Private_66:68:02	ARP	60	192.168.3.1 is at cc:01:48:2d:00:00
15	94.498732	192.168.3.30	192.168.4.40	ICMP	98	Echo (ping) request id=0x779b, seq=1/256, ttl=64 (reply in 16)
16	94.528006	192.168.4.40	192.168.3.30	ICMP	98	Echo (ping) reply id=0x779b, seq=1/256, ttl=63 (request in 15)
17	95.528532	192.168.3.30	192.168.4.40	ICMP	98	Echo (ping) request id=0x789b, seq=2/512, ttl=64 (reply in 18)
18	95.544180	192.168.4.40	192.168.3.30	ICMP	98	Echo (ping) reply id=0x789b, seq=2/512, ttl=63 (request in 17)
19	96.544539	192.168.3.30	192.168.4.40	ICMP	98	Echo (ping) request id=0x799b, seq=3/768, ttl=64 (reply in 20)
20	96.560446	192.168.4.40	192.168.3.30	ICMP	98	Echo (ping) reply id=0x799b, seq=3/768, ttl=63 (request in 19)
21	97.561526	192.168.3.30	192.168.4.40	ICMP	98	Echo (ping) request id=0xa9b, seq=4/1024, ttl=64 (reply in 22)
22	97.577397	192.168.4.40	192.168.3.30	ICMP	98	Echo (ping) reply id=0xa9b, seq=4/1024, ttl=63 (request in 21)
23	98.578609	192.168.3.30	192.168.4.40	ICMP	98	Echo (ping) request id=0xb9b, seq=5/1280, ttl=64 (reply in 24)
24	98.593253	192.168.4.40	192.168.3.30	ICMP	98	Echo (ping) reply id=0xb9b, seq=5/1280, ttl=63 (request in 23)

> Frame 14: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bit

> Ethernet II, Src: cc:01:48:2d:00:00 (cc:01:48:2d:00:00), Dst: Private_66:

- Address Resolution Protocol (reply)
 - Hardware type: Ethernet (1)
 - Protocol type: IPv4 (0x0800)
 - Hardware size: 6
 - Protocol size: 4
 - Opcodes: reply (2)
 - Sender MAC address: cc:01:48:2d:00:00 (cc:01:48:2d:00:00)
 - Sender IP address: 192.168.3.1
 - Target MAC address: Private_66:68:02 (00:50:79:66:68:02)
 - Target IP address: 192.168.3.30

```

0000  00 50 79 66 68 02 cc 01 48 2d 00 00 08 06 00 01    Pyfh... H.....
0010  08 00 06 04 00 02 c1 48 2d 00 00 c0 a8 03 01    .....H.....
0020  00 50 79 66 68 02 c0 a8 03 1e 00 00 00 00 00    Pyfh.....
0030  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00    ....

```

Затем PC3 формирует ICMP Echo Request (ping) с IP-адресом назначения 192.168.4.40. При этом IP-адрес назначения остаётся 192.168.4.40, а MAC-адрес назначения в Ethernet-кадре указывается MAC-адрес шлюза:

No.	Time	Source	Destination	Protocol	Length	Info
13	94.494516	Private_66:68:02	Broadcast	ARP	64	Who has 192.168.3.1? Tell 192.168.3.30
14	94.497768	cc:01:48:2d:00:00	Private_66:68:02	ARP	60	192.168.3.1 is at cc:01:48:2d:00:00
15	94.498732	192.168.3.30	192.168.4.40	ICMP	98	Echo (ping) request id=0x779b, seq=1/256, ttl=64 (reply in 16)
16	94.528006	192.168.4.40	192.168.3.30	ICMP	98	Echo (ping) reply id=0x779b, seq=1/256, ttl=63 (request in 15)
17	95.528532	192.168.3.30	192.168.4.40	ICMP	98	Echo (ping) request id=0x789b, seq=2/512, ttl=64 (reply in 18)
18	95.544180	192.168.4.40	192.168.3.30	ICMP	98	Echo (ping) reply id=0x789b, seq=2/512, ttl=63 (request in 17)
19	96.544539	192.168.3.30	192.168.4.40	ICMP	98	Echo (ping) request id=0x799b, seq=3/768, ttl=64 (reply in 20)
20	96.560446	192.168.4.40	192.168.3.30	ICMP	98	Echo (ping) reply id=0x799b, seq=3/768, ttl=63 (request in 19)
21	97.561526	192.168.3.30	192.168.4.40	ICMP	98	Echo (ping) request id=0x7a9b, seq=4/1024, ttl=64 (reply in 22)
22	97.577397	192.168.4.40	192.168.3.30	ICMP	98	Echo (ping) reply id=0x7a9b, seq=4/1024, ttl=63 (request in 21)
23	98.578609	192.168.3.30	192.168.4.40	ICMP	98	Echo (ping) request id=0x7b9b, seq=5/1280, ttl=64 (reply in 24)
24	98.593253	192.168.4.40	192.168.3.30	ICMP	98	Echo (ping) reply id=0x7b9b, seq=5/1280, ttl=63 (request in 23)

> Frame 15: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 b) on interface 0 Ethernet II, Src: Private_66:68:02 (00:50:79:66:68:02), Dst: cc:01:48:2d:00:00 (cc:01:48:2d:00:00) > Destination: cc:01:48:2d:00:00 (cc:01:48:2d:00:00) > Source: Private_66:68:02 (00:50:79:66:68:02) Type: IPv4 (0x0800) [Stream index: 3] > Internet Protocol Version 4, Src: 192.168.3.30, Dst: 192.168.4.40 0100 = Version: 4 0101 = Header Length: 20 bytes (5) > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT) Total Length: 84 Identification: 0x9b77 (39799) > 000. = Flags: 0x0 ...0 0000 0000 0000 = Fragment Offset: 0		<pre> 0000 cc 01 48 2d 00 00 00 50 79 66 68 02 08 00 45 00 ..H...P yfh...E... 0010 00 54 9b 77 00 00 40 01 56 9b c0 a8 03 1e c0 a8 T.w.@ V... 0020 04 28 08 00 a8 6f 77 9b 00 01 08 09 0a 0b 0c 0d (...ow..... 0030 0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d 0040 1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d ..!"#\$%&'()*+,-./:;=<?@A[\\]^_`{ }~. 0050 2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d ..123456789;=<?@A[\\]^_`{ }~. 0060 3e 3f </pre>
--	--	---

PC4 получает пакет и отправляет запрос IP маршрутизатора для получения его MAC-адреса:

```
Padding: 00000000000000000000000000000000
Frame check sequence: 0x00000000 [unverified]
[FCs Status: Unverified]
▼ Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: Private_66:68:03 (00:50:79:66:68:03)
    Sender IP address: 192.168.4.40
    Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
    Target IP address: 192.168.4.1
```

```
0000 ff ff ff ff ff ff 00 50 79 66 68 03 08 06 00 01 .....P yfh....
0010 08 00 06 04 00 01 00 50 79 66 68 03 c0 a8 04 28 .....P yfh... (
0020 ff ff ff ff ff ff c0 a8 04 01 00 00 00 00 00 00 .....
0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
```

arp icmp						
No.	Time	Source	Destination	Protocol	Length	Info
12	94.497799	192.168.3.30	192.168.4.40	ICMP	98	Echo (ping) request id=0x779b, seq=1/256, ttl=63 (reply in 15)
13	94.497961	Private_66:68:03	Broadcast	ARP	64	Who has 192.168.4.1? Tell 192.168.4.40
14	94.507839	cc:01:48:2d:00:10	Private_66:68:03	ARP	60	192.168.4.1 is at cc:01:48:2d:00:10
15	94.508613	192.168.4.40	192.168.3.30	ICMP	98	Echo (ping) reply id=0x779b, seq=1/256, ttl=64 (request in 12)
16	95.524039	192.168.3.30	192.168.4.40	ICMP	98	Echo (ping) request id=0x789b, seq=2/512, ttl=63 (reply in 17)
17	95.524150	192.168.4.40	192.168.3.30	ICMP	98	Echo (ping) reply id=0x789b, seq=2/512, ttl=64 (request in 16)
18	96.540333	192.168.3.30	192.168.4.40	ICMP	98	Echo (ping) request id=0x799b, seq=3/768, ttl=63 (reply in 19)
19	96.540447	192.168.4.40	192.168.3.30	ICMP	98	Echo (ping) reply id=0x799b, seq=3/768, ttl=64 (request in 18)
20	97.557231	192.168.3.30	192.168.4.40	ICMP	98	Echo (ping) request id=0x7a9b, seq=4/1024, ttl=63 (reply in 21)
21	97.557362	192.168.4.40	192.168.3.30	ICMP	98	Echo (ping) reply id=0x7a9b, seq=4/1024, ttl=64 (request in 20)
22	98.573891	192.168.3.30	192.168.4.40	ICMP	98	Echo (ping) request id=0x7b9b, seq=5/1280, ttl=63 (reply in 23)
23	98.573227	192.168.4.40	192.168.3.30	ICMP	98	Echo (ping) reply id=0x7b9b, seq=5/1280, ttl=64 (request in 22)